Name: Key

8. a) Compute the determinant of this matrix:

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ -1 & 2 & 2 & 5 \\ 0 & 1 & 2 & 0 \\ 0 & 1 & 8 & 0 \end{pmatrix}$$

By cofactor expansion along 4th row:

$$\det A = +1 \cdot \begin{vmatrix} 2 & 3 & 4 \\ -1 & 2 & 5 \\ 0 & 2 & 0 \end{vmatrix} - 8 \begin{vmatrix} 2 & 1 & 4 \\ -1 & 2 & 5 \\ 0 & 1 & 0 \end{vmatrix}$$

Again by cofactor expansion:

$$= 1 \cdot \left(-2 \cdot \begin{vmatrix} 2 & 4 \\ -1 & 5 \end{vmatrix}\right) - 8\left(-1 \cdot \begin{vmatrix} 2 & 4 \\ -1 & 5 \end{vmatrix}\right)$$

$$= -2(10-(-4)) + 8(10+4) = 6 \cdot 14 = 84$$

b) Give two examples A and B of 2×2 matrices with determinant -5. Then using your examples, calculate $\det(AB)$ and $\det(3A)$.

For A and B, many examples are possible:

e.g. $A = \begin{pmatrix} -5 & 0 \\ 0 & 1 \end{pmatrix}$ $\quad B = \begin{pmatrix} 0 & 5 \\ 1 & 0 \end{pmatrix}$

$\det(AB) = \det(A) \cdot \det(B) = (-5)(-5) = \boxed{25}$

$\det(3A) = 9 \det A = 9 \cdot (-5) = \boxed{-45}$

by multilinearity, since A is 2×2.

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By cofactor expansion along 4th row:

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Again by cofactor expansion:

$$= 1\left(-2 \cdot \begin{vmatrix} 2 & 2 \\ -1 & 5 \end{vmatrix}\right) - 8\left(-1 \cdot \begin{vmatrix} 2 & 2 \\ -1 & 5 \end{vmatrix}\right)$$

$$= -2(10+2) + 8(10+2) = 6 \cdot 12 = 72$$

b) Give two examples A and B of 2×2 matrices with determinant -2. Then using your examples, calculate $\det(AB)$ and $\det(A^3)$.

For A and B, many examples are possible.

e.g. $A = \begin{pmatrix} 5 & 4 \\ 3 & 2 \end{pmatrix}$ $B = \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix}$

$\det(AB) = \det(A)\det(B) = (-2)(-2) = \boxed{4}$

$\det(A^3) = (\det A)^3 = (-2)^3 = \boxed{-8}$

For A and B, many examples are possible:

A= $\begin{pmatrix} 5 & 4 \\ 3 & 2 \end{pmatrix}$ **B=** $\begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix}$

det(AB) = det(A)det(B) = (-2)(-2) = 4

det(A^3) = (det(A))^3 = (-2)^3 = -8